

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Conservation Agriculture to One Million Hectares in Morocco: Building Climate Resilience and Employment for Rural Youth

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.1 · Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

Morocco's agriculture is at a critical juncture. Facing structural drought, severe rainfall variability, and a national imperative to enhance rural livelihoods, the government's "Generation Green 2020–2030" strategy aims to scale Conservation Agriculture (CA) to one million hectares by 2030. This Scaling Solution Track (SST) addresses the core challenges preventing the acceleration of CA adoption: limited farmer awareness, restricted access to scale-appropriate no-till seeders and adapted seeds, and the need for robust policy support and institutional incentives. The primary goal is to support the Moroccan government and its partners in achieving this ambitious national target, thereby building climate resilience, improving food security, and creating new economic opportunities for rural youth and women.

This SST will operate across Morocco's rainfed cereal zones, where over 80% of production is vulnerable to climate shocks. The solution is a comprehensive innovation bundle that integrates climate-smart, CA-adapted crop varieties (cereals and legumes); locally manufactured no-till seeders; a digital platform (Zra3Direct) for advisory and seeder-hire services; and plot-specific agronomic guidance. This is not merely a technology transfer project; it is a systemic scaling effort designed to strengthen the entire value chain—from seed multiplication and mechanization to last-mile service delivery and policy influence.

The track is built on over 30 years of collaborative research between ICARDA and Morocco's National Institute for Agricultural Research (INRA). Evidence from farmer-managed trials and large-scale adoption (CA area has grown from 10,500 ha in 2015 to 160,000 ha in 2024/25) confirms the package's effectiveness in increasing yields by 40–60% in drought years, and by 10–20% with net profit gains of approximately 21% in normal rainfall seasons. Core implementing partners include the Ministry of Agriculture (MAPMDREF), INRA, extension agencies (ONCA), farmers associations for the adoption of conservation agriculture (AMAC), farmers association for the multiplication of suitable seeds (AMMS), private sector carbon farming partners (Tourba), and private seed (SONACOS) and machinery companies (ATMAR, and others).

2. Innovation Bundle & Scaling Ambition

2.1 What is Being Scaled

- **Climate-Smart Seeds:** Drought-tolerant, early-maturing varieties of durum wheat, bread wheat, barley, lentil, chickpea, and faba bean, bred for CA systems. ICARDA and INRA jointly produce early-generation seed (G1/G2) at the Marchouch Research Station, which is then multiplied through SONACOS and private seed companies for farmer distribution.
- **No-Till Seeders:** Access to low-cost, locally manufactured no-till seeders through youth-led service provision and government-subsidized ownership.
- **Digital Advisory Platform (Zra3Direct):** A platform that connects farmers with seeder service providers, tracks CA adoption at national level, and provides agronomic advice. Zar3Direct is recognised by the Ministry of Agriculture as a core monitoring tool for the one million hectare initiative.
- **CA-Based Agronomic Package:** Integrated nutrient, pest, weed, and crop residue management practices, including cereal-legume rotation.

2.2 Evidence of Impact

- **Yield Resilience:** In severe drought years, the full CA package (improved variety + zero till) increases cereal yields by **40–60%** compared to conventional tillage.
- **Agronomic Gains:** In normal seasons, CA practices increase yields by **10–20%** and net profits by approximately **21%**.

- **Soil Health:** CA enhances soil fertility, moisture retention, and reduces erosion, supporting long-term sustainability and carbon sequestration. ICARDA and INRA's published soil carbon research underpins the Verra-certified MRV methodology used by Tourba for carbon credit generation under CA.
- **Adoption Momentum:** National CA area has increased more than **15-fold** in nine years, from 10,500 ha (2014/15 season) to 160,000 ha declared (2024/25 season), demonstrating strong national demand and scalability.

2.3 Scaling Targets

Area under CA practices (cumulative)	250,000 ha		1,000,000 ha	
Certified seed production	CA-adapted	Support 20+ ha	early-generation seed on	1,000 quintals of G2 from recent varieties. 1,000,000 quintals of certified seeds of national varieties sold
No-till seeders available	Support distribution	government's (~235)	annual	1,000 planters in operation

3. Scaling Pathway & Theory of Change

3.1 Pathway Strategy The scaling pathway is a public-private-philanthropic partnership (PPPP) model, deeply integrated into Morocco's national governance structures. The national CA steering committee — chaired by INRA with secretariat support and comprising the Ministry of Agriculture's regional directorates, extension agencies, seed and machinery companies, farmer associations (AMAC), seed companies (SONACOS) and the multipliers that serve it (AMMS), and the private-sector carbon farming partner Tourba — provides the governance architecture for coordinated scaling across regions.

First, the pathway focuses on **demand creation and capacity building** through on-farm demonstrations, field days, and participation in major national events like the International Agricultural Fair of Morocco (SIAM). Second, it strengthens **last-mile delivery** by supporting youth-led service provision for mechanization and leveraging the Zra3Direct digital platform to connect farmers with service providers. Third, it fosters **systemic change** by providing scientific backstopping to the national steering committee, guiding policy on subsidies and seed certification (ONSSA), and linking farmers to carbon market opportunities. This approach ensures that scaling is not a linear push but a coordinated, multi-stakeholder effort.

3.2 Theory of Change

If smallholder farmers in Morocco gain reliable access to a bundled package of climate-smart seeds, no-till seeders (via youth-led services), and CA-based agronomic advice, **and** if key market and coordination failures—such as early-generation seed bottlenecks, seeder repair ecosystems, and knowledge gaps—are addressed through public-private partnerships and strengthened regional steering committees, **then** farmers will adopt and sustain CA practices, leading to increased yields, stabilized incomes, and enhanced climate resilience, **ultimately contributing** to the national goal of one million hectares under CA, the creation of rural jobs, and a more sustainable and food-secure agricultural system in Morocco.

3.3 Key Activities Funded by S4I

- **Scientific Backstopping & Policy Guidance:** Informing the Ministry of Agriculture's CA roadmap, guiding suitable machinery deployment, and supporting policy briefs.
- **Early Seed System Support:** Producing early-generation (G1/G2) seeds of new CA-adapted varieties to break the bottleneck in the multiplication chain.
- **Digital Platform Development:** Enhancing the Zra3Direct platform for adoption monitoring, service provider coordination, and integration of new functionalities including residue deposition and crop rotation tracking.
- **Demand Creation & Awareness:** Co-organizing targeted campaigns, field days, and participation in the SIAM agricultural fair.

4. Partners & Roles

Partner Type	Specific Partners & Quantities	Role in Scaling Pathway
Demand Partners	• Associations: AMAC (Moroccan CA Association), COMADER • Private Sector: SONACOS (seed, 96% market share), AMMS (seed), ATMAR, SOLA and other machinery manufacturers • Financial: Crédit Agricole, ADA (rural development agency)	Articulate farmer needs, drive demand for seeds and mechanization services, invest in and maintain equipment, provide last-mile rental services, and adopt CA practices.
Scaling Partners	• Government: MAPMDREF (Ministry of Agriculture), DPA (Provincial Directorates) • Research & Extension: INRA (National Research), ONCA (extension), Tourba, Al Moutmir (extension entities), ONSSA (seed certification) • Development Programs: World Bank, IDRC (Wheat Accelerator), FAO, African Union	Co-lead national and regional steering committees, set and implement national policy (Generation Green), provide subsidy frameworks, conduct research, multiply and certify seeds, and deliver extension training.

4.4 Funding Model: S4I catalytic funding provides scientific and technical support to guide policy, de-risk early seed production, and enhance digital tools, while the Government of Morocco finances core subsidies (50% for seeders, certified seeds) and international partners (World Bank, IDRC) fund complementary research and carbon initiatives.

5. Enabling Environment & Risks

5.1 Opportunities

- **Strong Government Ownership:** CA adoption is a formal target of the "Generation Green 2030" strategy, with an established national steering committee, and a clear national roadmap.
- **Existing Scaling Momentum:** CA adoption has surpassed 160,000 ha, with functional partnerships, local no-till seeder manufacturers, a robust seed certification system, and an emerging carbon market linkage.
- **High-Level Political Will:** Drought is a structural reality, creating a powerful political and farmer-driven pull for climate-resilient solutions like CA.

- **Leverageable Investments:** Complementary programs from the World Bank, IDRC, and FAO provide additional resources, creating a strong foundation for S4I additionality.

5.2 Challenges & Mitigation

- **Mechanization Supply Capacity:** Reaching one million hectares will require sustained growth in the national seeder fleet.
- *Mitigation:* Support the national CA steering committee to source and coordinate annual seeder allocation across regions to optimize coverage.
- **Early-Generation Seed Bottleneck:** A lack of nationally produced G1/G2 seed of adapted varieties leads to reliance on imported seeds.
- *Mitigation:* S4I funding will directly support the multiplication of early-generation seeds over 30 ha to kickstart the national supply chain.
- **Adoption & Behavior Change:** CA is knowledge-intensive, and farmers are risk-averse, especially during drought years. Competition for crop residues with livestock is a deep-seated challenge.
- *Mitigation:* Use a combination of intensive on-farm demonstrations, peer-to-peer exchanges, and digital tools (Zra3Direct) for continuous learning. Explicitly address residue management in training and explore integrated crop-livestock solutions.

• 6. Learning & Adaptation

6.1 Monitoring Focus

- **Process Outcomes:** Functionality of the national and regional CA steering committees; number and effectiveness of public-private partnerships formed.
- **Behavior Change:** Shifts in farmer awareness and demand for CA; utilization rates of no-till seeders; youth and women's participation in service provision.
- **System Outcomes:** Volume of early-generation and certified seed produced; availability and affordability of spare parts; evidence of policy decisions influenced by project-generated evidence.
- **S4I Additionality:** Clear documentation of outcomes directly linked to S4I-funded activities (e.g., tons of G1 and G2 seed produced, policy briefs adopted, and training events delivered).

6.2 Adaptive Management

- **Quarterly Check-ins:** Regular meetings between the ICARDA team and the S4I Area of Work 2 team to review progress against milestones and discuss emerging challenges.
- **Annual "Pause & Reflect":** A formal workshop with all key partners to synthesize lessons learned, evaluate progress, and agree on adaptive measures for the following year.
- **Steering Committee Integration:** Lessons will be channeled directly into the national CA steering committee to inform real-time decision-making.

7. Next Steps & Timeline (2026)

Quarter	Key Activities
Q1	Finalize detailed work plan and budget based on 2024/25 CA adoption data (160,000 ha). Initiate early-generation seed multiplication activities. Begin planning for SIAM agricultural fair participation (April).
Q2	Lead ICARDA/S4I presence at SIAM fair for demand creation and high-level engagement. Conduct first round of on-station demonstrations and field days.
Q3	Convene workshops with regional CA steering committees to review progress and refine scaling strategies. Begin data collection for mid-year KPIs. Support youth service provider training.
Q4	Support no-till seeding and other conservation agriculture activities for the start of the 2026/27 season. Consolidate end-of-year reports on adoption and KPIs

Communications & Knowledge: Key products will include annual progress reports, policy briefs for the Ministry of Agriculture, and success stories highlighting youth and women service providers. Channels will include the CGIAR website, participation in SIAM, and direct engagement with national media (e.g.,).

8. Contact

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